

EXCEL SCATTER CHART

Explore relationships between two numbers

10

WHEN TO USE

- Data has **two numeric variables** measured for each record.
- Answers: **Do the two variables move together?**
- Use it to spot **correlation, patterns and outliers** before deeper analysis.

BUSINESS QUESTION

- Does higher advertising spend lead to higher sales?
- Is the relationship positive, negative or none?
- Which points are outliers worth investigating?

DATA EXAMPLE

	A	B
1	Ad Spend	Sales
2	5	40
3	8	52
4	10	58
5	14	70
6	17	81
7	20	88
8	24	101
9	28	64

DATA ANALYST USE CASE

Example: **Marketing Analysis**

QUESTION

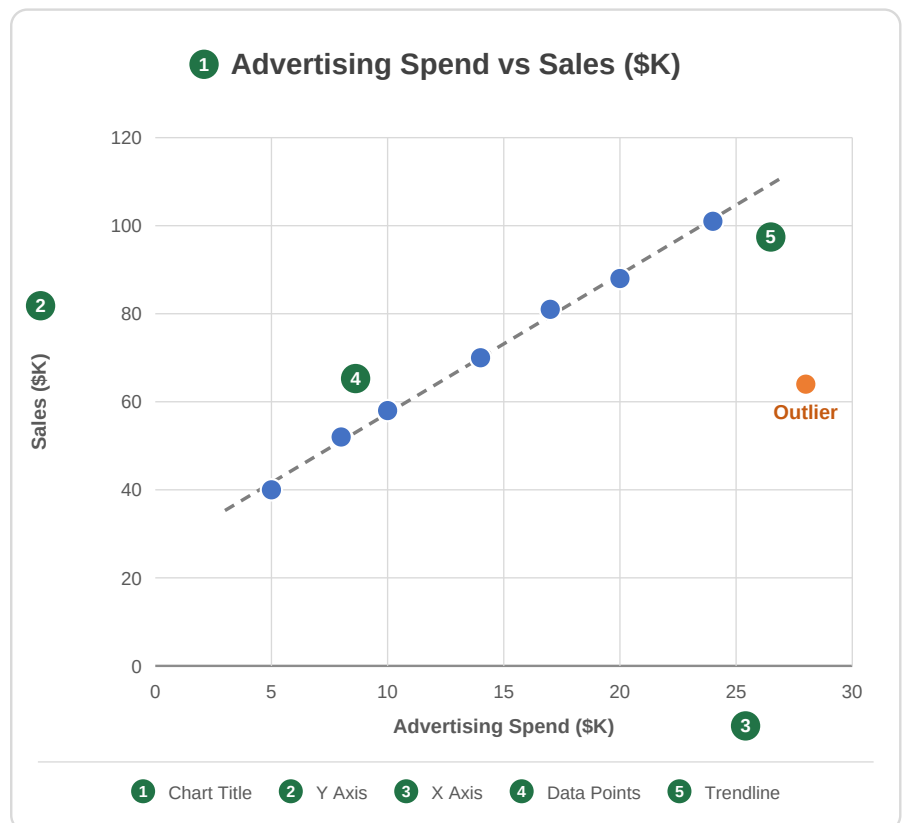
Does advertising drive sales?

CHART

Scatter Chart

INSIGHT

Sales rise steadily with ad spend, except one outlier at 28K spend (64K sales) worth investigating.



HOW TO CREATE IT IN EXCEL



- 1** Select the two numeric columns, including the headers (A1:B9).
- 2** Go to the **Insert** tab and open the **Charts** group.
- 3** Click **Insert Scatter (X, Y) or Bubble Chart > Scatter** (markers only).
- 4** Add a **Chart Title** and **axis titles** for both variables, with units.
- 5** **Format**: add a **Trendline** (Chart Elements > Trendline) and investigate outliers.

DO / DON'T

DO ✓

- ✓ Put the driver (X) on the horizontal axis
- ✓ Label both axes with units
- ✓ Add a trendline
- ✓ Investigate outliers

DON'T ✗

- ✗ Use it for categories
- ✗ Assume correlation means causation
- ✗ Connect the points with lines
- ✗ Ignore outliers



DATA ANALYST TIP

Use it to explore relationships between two numerical variables.

Correlation is not causation. Use scatter charts to find questions, not final answers.